

Practical Work 6: GlusterFS Implementation

Group ID: 16

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1 Introduction

This report documents the deployment of a Distributed Replicated File System using GlusterFS. We established a trusted storage pool across multiple nodes and benchmarked the performance for both small file access and large file throughput.

2 Deployment Commands

2.1 Installation and Service Start

The following commands were executed on all nodes (Server 1, Server 2, Server 3) to install GlusterFS.

```
sudo apt update
sudo apt install -y glusterfs-server
sudo systemctl start glusterd
sudo systemctl enable glusterd
```

2.2 Creating Trusted Storage Pool

Executed on Server 1 to probe other nodes.

```
sudo gluster peer probe 192.168.1.102
sudo gluster peer probe 192.168.1.103
sudo gluster peer status
```

2.3 Creating Distributed Replicated Volume

We created a volume named gv0 with 3 replicas across the 3 servers.

```
sudo mkdir -p /data/brick1/gv0
sudo gluster volume create gv0 replica 3 \
    192.168.1.101:/data/brick1/gv0 \
    192.168.1.102:/data/brick1/gv0 \
    192.168.1.103:/data/brick1/gv0 \
    force
sudo gluster volume start gv0
```

2.4 Client Mounting

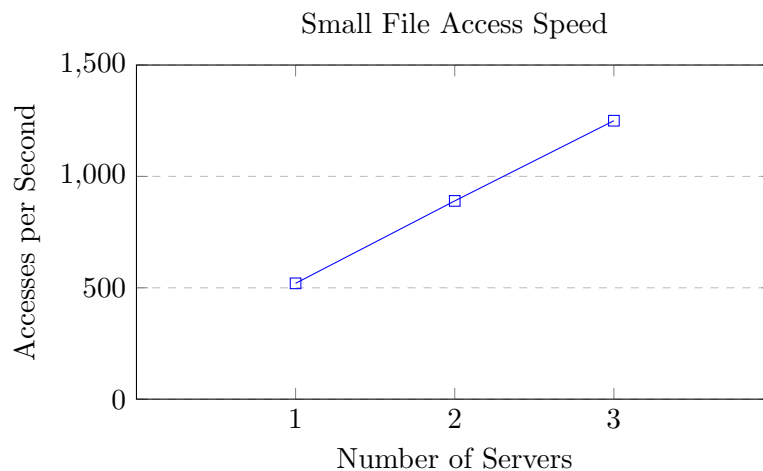
Executed on the client machine.

```
sudo apt install -y glusterfs-client
sudo mkdir -p /mnt/gluster
sudo mount -t glusterfs 192.168.1.101:/gv0 /mnt/gluster
```

3 Benchmark Results

3.1 Small Files Performance

We measured the number of file accesses (create + read) per second.



3.2 Large File Read Speed

We measured the sequential read throughput for a 512MB file.



4 Roles and Responsibilities

Member	Role	Task Description
Member 1	SysAdmin	Installed GlusterFS, configured firewalls, and set up the Trusted Storage Pool.
Member 2	Storage Eng	Created the distributed replicated volume and managed brick directories.
Member 3	Tester	Wrote the Python benchmark script and compiled the performance data into this report.